

Member contribution plan

IT3012 - Intelligent Agents

Group 2026-AI-45 | Finalized 10 October 2026

Confirmed ownership for Q1-Q7

The team has finalized an 8 / 12 / 12 / 8 allocation of responsibility for the 40 group implementation marks. These totals describe assigned scope, not individual grades or achieved scores.

Member	Name / submission spelling	Registration	GitHub username
Member 1	Adithya Gunawardana Gunawardana M.A.A.	IT24103038	AdithyaGunawardana
Member 2	Ushan Srinuka Srinuka D.G.U.	IT24100184	Ushan-Srinuka
Member 3	Sanuda Abeysinghe Abeysinghe S.D.	IT24100788	sanudaabey
Member 4	Wanshaja Sooriyabandara Sooriyabandara U.R.G.W.K.	IT24102798	WKS2004

Allocation

Owner	Questions	Marks	Estimated workload
Adithya	Q1: DFS + Q3: UCS	8	Moderate
Ushan	Q2: BFS + Q5: CornersProblem	12	Moderate-High
Sanuda	Q4: A* + Q6: Corners heuristic	12	High
Wanshaja	Q7: Food heuristic	8	High

BFS and the corner problem stay with Ushan; A* and the corner heuristic stay with Sanuda. Wanshaja has dedicated time for the food heuristic. Adithya coordinates integration and verification alongside the two generic searches.

Status: responsibilities finalized. Implementation, grading screenshots and completed-work evidence remain pending. Member numbering follows this project, not OCEAVERA.

Question deliverables

Each owner implements, tests and explains their questions. Supplied interfaces, framework files and grading expectations must be preserved.

Question	Owner	Main implementation
Q1 4 marks	Adithya	depthFirstSearch Stack-based graph search; track expanded states and return legal action paths.
Q2 4 marks	Ushan	breadthFirstSearch Queue-based graph search; shortest paths by steps when costs are equal.
Q3 4 marks	Adithya	uniformCostSearch Priority g; accumulated costs, cheaper-path handling and stale entries.
Q4 4 marks	Sanuda	aStarSearch Priority g + h; use the heuristic argument. Zero heuristic behaves as UCS.
Q5 8 marks	Ushan	CornersProblem Small hashable position/visited-corners state; start, goal and legal successors.
Q6 8 marks	Sanuda	cornersHeuristic Admissible and consistent estimate; nonnegative and zero at corner goals.
Q7 8 marks	Wanshaja	foodHeuristic Admissible and consistent food estimate; tune performance and cache where useful.

Heuristic acceptance targets

Question	PDF full-credit benchmark	Correctness requirement
Q6	mediumCorners: at most 1,200 expanded nodes	Admissible, consistent and optimal. An inadmissible heuristic or detected non-optimal path receives zero.
Q7	trickySearch: at most 9,000 expanded nodes	Admissible and consistent; retain correct optimal search behavior.

Q1-Q4: existing src/search.py. Q5-Q7: existing src/searchAgents.py. All other source files are read-only. Use util.Stack, util.Queue and util.PriorityQueue for the fringes. Q8 is outside scope.

Node limits are targets, not reported results. The local grader's raw points differ from the PDF's assignment marks.

Handoffs, evidence and submission

Coordinator	Supporting responsibility
Adithya	Integrate changes and coordinate final Q1-Q7 verification; compare UCS and A* behavior.
Ushan	Assemble the report and specify the corner-state format for Sanuda before Q6 implementation.
Sanuda	Coordinate A*/Q6 integration and heuristic defense; supply working A* for Wanshaja's Q7 testing.
Wanshaja	Prepare food performance evidence and consolidate genuine Git contribution evidence.
Everyone	Make meaningful commits, provide owned-question evidence, review peers and understand the entire solution for the six-minute team viva.

Evidence each owner supplies

Identify edited functions/code blocks, capture actual autograder output and explain the question in at most 200 words. Heuristic owners also supply correctness arguments and actual expansion counts. Keep results and screenshots outside src/; run Python with -B.

Final report prerequisites

Complete Q1-Q7, verify required behavior, collect all seven grading screenshots, reconcile actual contributions with Git history, capture contribution graphs and complete the AI declaration with exact prompts from all project assistance.

Required submission files

Group_2026-AI-45_Report.pdf - completed standalone report.

Group_2026-AI-45_Code.zip - only search.py and searchAgents.py at the ZIP root. Submit once per group.

This contribution plan does not replace the submission report. The public-repository requirement conflicts with the supplied source's solution-publication restriction; obtain instructor clarification before publishing completed solutions.

Identity sources and privacy

Registration numbers and formal names: OCEAVERA initial submission, 2026-AI-45-Machine_Learning-Initial_Submission.pdf, pages 1-2. GitHub usernames: OCEAVERA docs/project/ai-team-members.md, cross-checked against this repository's contributor list. The initial PDF contains no GitHub account links.

Only names, registration numbers and GitHub usernames are included as member identity data. Canonical editable documents: docs/member-contributions.md and docs/report-submission-plan.md.